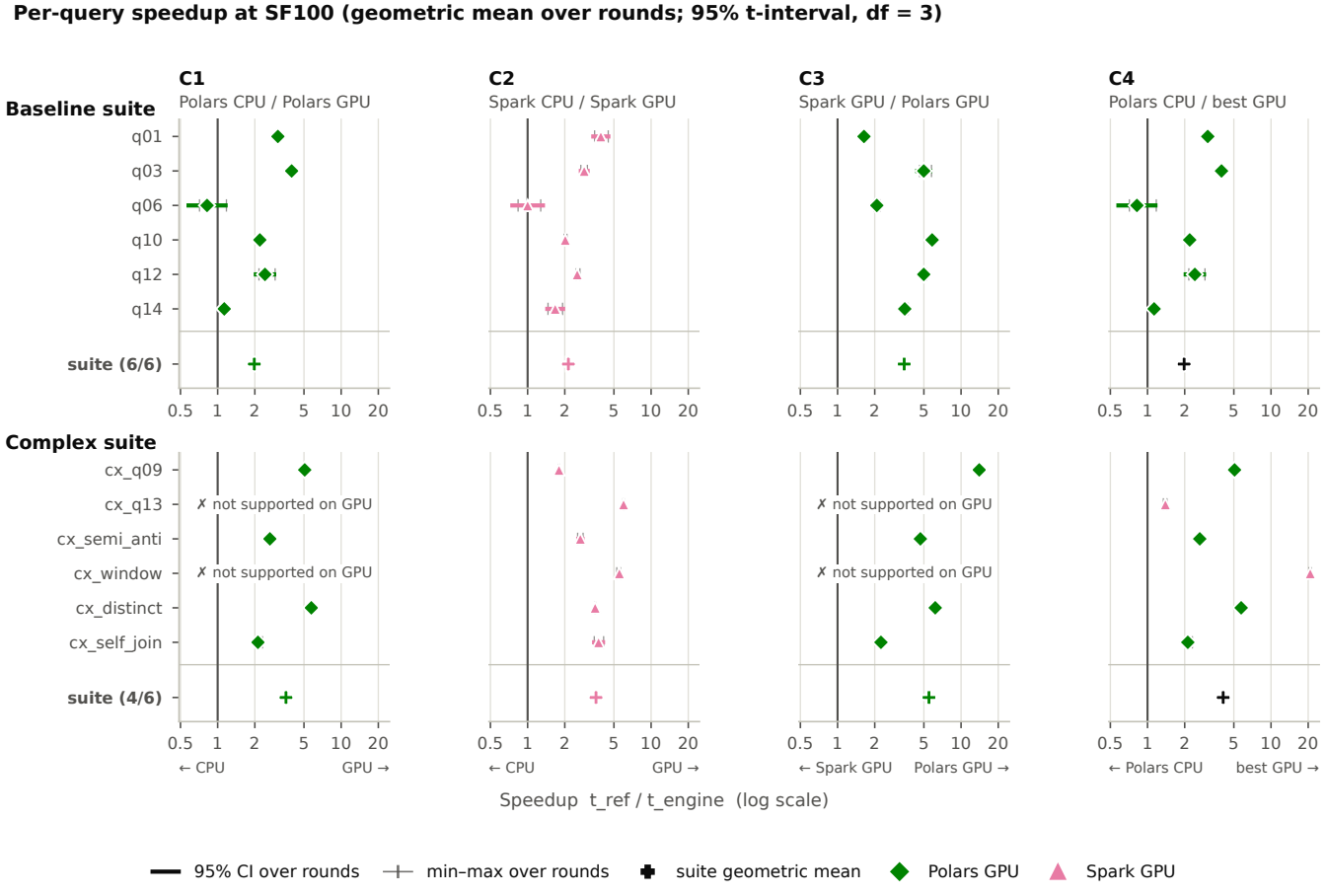
Per-query speedup at SF100 (geometric mean over rounds; 95% t-interval, df = 3)
C1
Polars CPU / Polars GPU
C2
Spark CPU / Spark GPU
C3
Spark GPU / Polars GPU
C4
Polars CPU / best GPU
Baseline suite
q01
q03
q06
q10
q12
q14
suite (6/6)
Complex suite
cx_q09
cx_q13
cx_semi_anti
cx_window
cx_distinct
cx_self_join
suite (4/6)
✗ not supported on GPU
0.5 1 2 5 10 20
← CPU
GPU →
← Spark GPU
Polars GPU →
← Polars CPU
best GPU →
Speedup t_ref / t_engine (log scale)
95% CI over rounds
min–max over rounds
suite geometric mean
Polars GPU
Spark GPU